How Autocracy Shapes the Social Sciences: Synthesized Evidence from an Agent Orchestra

Pre-analysis draft — not for circulation

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Abstract

[Does authoritarian rule constrain the production of social science and humanities (SSH) knowledge? We test 25 pre-registered hypotheses across five behavioral families using a corpus of ~2.7 million SSH articles from the Web of Science (1970–2023), matched to V-DEM country-year democracy scores. Each hypothesis is estimated by a dedicated analyst agent using two-way fixed effects regression. Preliminary results: [TODO: fill after analysis.]]

Contents

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1 Introduction

[Motivating question: How does autocracy constrain what social scientists study, how they frame it, who they collaborate with, and how visible their work becomes?]

Contribution.

- Pre-registered multi-hypothesis test of self-censorship theory
- 25 teams, 5 behavioral families, Bonferroni correction within families
- Agent-orchestra methodology: reproducible, scalable, transparent
- Corpus: 2.7M SSH articles, 1970–2023, 167 countries

Roadmap. *[Section outline.]*

2 Theory: Self-Censorship under Autocracy

[Core theoretical claim: researchers in autocracies face career costs for politically sensitive topics, methods, collaborations, and framings. This produces five families of observable behavioral adaptations.]

2.1 Mechanisms

1. **Topic avoidance** — steer away from politically sensitive subjects
2. **Framing neutrality** — hedge, depoliticize, adopt technocratic language
3. **Collaboration constraint** — avoid international/liberal-democratic partners
4. **Visibility suppression** — reduced citation uptake of self-censored output
5. **Ideological alignment** — active production of regime-consonant framing

2.2 Bonferroni families and thresholds

Table 1 lists the five theory sub-families, the teams assigned to each, and the Bonferroni-corrected significance threshold.

Table 1: Theory families, team assignments, and Bonferroni thresholds

Sub-family	Teams	k	Adjusted α ($0.05/k$)
topic-avoidance	01–07	7	$p < 0.0071$
framing-neutrality	08–13	6	$p < 0.0083$
collaboration-constraint	14–18	5	$p < 0.0100$
visibility-suppression	19–22	4	$p < 0.0125$
ideological-alignment	23–25	3	$p < 0.0167$
Total	01–25	**25**	

3 Data and Corpus

3.1 Bibliometric corpus (`agent_corpus.rds`)

- Source: Web of Science (WOS), Social Sciences and Humanities categories
- Raw rows: 3,189,557 (multi-author; one row per author-article pair)
- Unique articles (`wos_id`): 2,709,224
- Years covered: 1970–2023
- Country match rate: 99.98% (by `iso3` affiliation)
- Fields: all WOS SSH subject categories (see `ssh_fields.txt`)

3.2 Political data

- Primary IV: `v2x_libdem` (V-DEM Liberal Democracy Index, 0–1)
- Alternative IV: `lied_binary` (dichotomous democracy indicator)
- Controls: `e_gdppc` (GDP per capita), `e_wb_pop` (population)
- Regime category: `v2x_regime` (0–3, for heterogeneity tests)

3.3 Data preparation

- Script: `scripts/00_prepare_data.R` (de-duplicates, merges V-DEM)
- Country-year controls added: `scripts/add_controls.R`
- Country-year minimum: 10 distinct articles (most teams)
- Pre-registered corpus: locked before any analysis

4 Agent-Orchestra Methodology

4.1 Pipeline overview

[The multi-agent pipeline consists of five sequential agent types. Each team is assigned one instance of each agent. All designs are pre-registered before any analysis is run.]

Figure ?? illustrates the full pipeline.

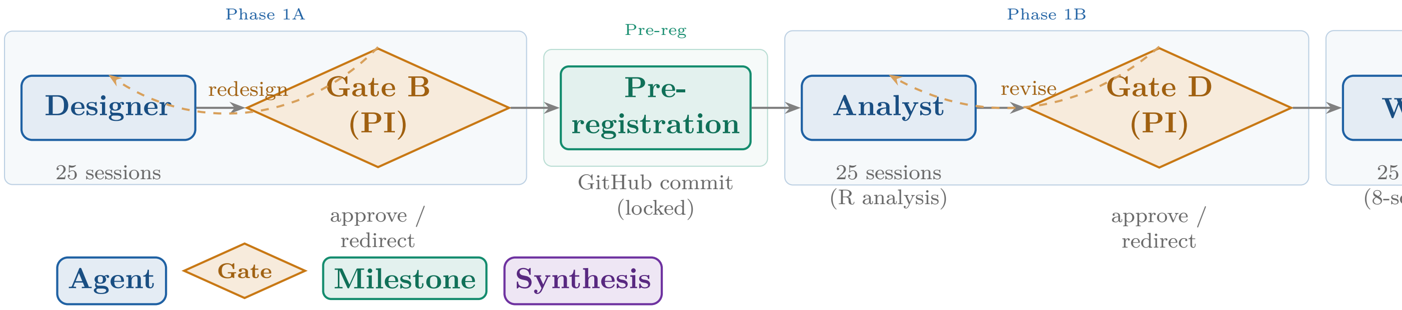

Figure 1: Agent-orchestra pipeline. **Blue**: agent sessions (Claude Sonnet/Haiku); **amber**: PI gate reviews; **teal**: pre-registration milestone; **violet**: synthesis. Dashed arrows indicate redesign/revision loops possible at each gate. Current status: pre-registration complete (gate B passed); analyst sessions pending.

4.2 Agent roles

Designer Develops the research question, operationalization, and analysis plan within a pre-assigned theoretical domain. Output: `rq.md` + `analysis_plan.md`.

Analyst Executes the pre-registered analysis in R using the locked corpus. Output: `analysis.R`, figures, `primary_results.json`.

Writer Drafts the team report following a standardized 8-section template. Output: `report.md`.

Reviewer Peer-reviews the report against rubric criteria. Output: `peer_review.md`.

Synthesizer Reads all 25 reports and writes the cross-team theory evaluation. Output: `synthesis/theory_evaluation.md`.

4.3 Pre-registration

All 25 teams' hypotheses and analysis plans were committed to GitHub (commit `960525f`) on 6 May 2026, before any analyst session was run. The timestamp and Git commit hash constitute the pre-registration record. Bonferroni adjustment is computed programmatically post-analysis using `scripts/bonferroni_adjust.R`.

4.4 Estimation strategy

Primary specification (all teams, country-year level). Two-way fixed effects (TWFE):

$$Y_{it} = \beta \cdot \texttt{v2x\_libdem}_{it} + \gamma_1 \log(\texttt{gdppc}_{it}) + \gamma_2 \log(\texttt{pop}_{it}) + \alpha_i + \delta_t + \varepsilon_{it} \tag{1}$$

where α_i = country FE, δ_t = year FE, SE clustered by `iso3`.

Secondary specification (pooled OLS). Year FE only (no country FE), SE clustered by `iso3`. Checks whether the within-country estimate is consistent with the cross-sectional pattern.

Robustness checks. All teams pre-specify RC1 (sample restriction or alternative operationalization) and RC2 (replace `v2x_libdem` with `lied_binary`).

5 Pre-registered Hypotheses

Table **??** lists all 25 pre-registered hypotheses with outcome variable, expected direction of the `v2x_libdem` coefficient, and additional controls beyond the baseline.

Table 2: Pre-registered hypotheses, all 25 teams

T	Outcome variable	H1 (condensed)	Sign on $\hat{\beta}$	Extra controls / notes	Type
Sub-family: topic-avoidance $k = 7$, $\alpha^* = 0.0071$					
01	`pci_mean` Mean PCI score (political-content keyword share of abstract)	Higher libdem → higher political-content keyword share	$\beta > 0$	—	Simple
02	`keyword_entropy_cy` Shannon entropy of author keyword distribution	Higher libdem → higher topical breadth of keyword vocabulary	$\beta > 0$	—	Simple
03	`share_sensitive_fields` Share of output in sensitive WOS disciplines	Higher libdem → higher share in politically sensitive fields (PolSci, IR, Law, Soc, etc.)	$\beta > 0$	—	Simple
04	`share_sensitive` Share of articles with regime-sensitive terms	Higher libdem → higher share mentioning democracy, rights, corruption, protest	$\beta > 0$	—	Simple
05	`share_critical_framing` Share critically examining own-country governance (LLM)	Higher libdem → higher share critically examining own-country institutions	$\beta > 0$	—	Complex
06	`mean_semantic_diversity` Mean pairwise cosine distance of embeddings	Higher libdem → higher semantic diversity within country-field-year clusters	$\beta > 0$	—	Complex

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T	Outcome variable	H1 (condensed)	Sign on $\hat{\beta}$	Extra controls / notes	Type
07	`share_intl_funded` Share acknowledging international funding agencies	Higher libdem → higher share with international funding acknowledgement	$\beta > 0$	—	Simple
Sub-family: framing-neutrality $k = 6$, $\alpha^* = 0.0083$					
08	`hedging_rate` Mean epistemic hedging phrases per abstract word	Higher libdem → lower hedging rate (autocracies hedge more)	$\beta < 0$	—	Simple
09	`normative_rate` Mean normative/prescriptive terms per abstract word	Higher libdem → higher normative language rate	$\beta > 0$	—	Simple
10	`share_technocratic` Share purely technocratic abstracts (LLM)	Higher libdem → lower share of exclusively technocratic abstracts	$\beta < 0$	—	Complex
11	`stance_rate` Rate of 1st-person argumentative stance phrases	Higher libdem → higher first-person argumentative stance rate	$\beta > 0$	—	Simple

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T	Outcome variable	H1 (condensed)	Sign on $\hat{\beta}$	Extra controls / notes	Type
12	`share_normative_conclusion` Share with explicit normative/policy conclusion (LLM)	Higher libdem → higher share of abstracts ending with normative conclusion	$\beta > 0$	—	Complex
13	`share_domestic_journal` Share published in domestically dominated journals	Higher libdem → lower share in domestically dominated journals	$\beta < 0$	—	Medium
Sub-family: collaboration-constraint $k = 5$, $\alpha^* = 0.0100$					
14	`mean_n_coauthor_cntry` Mean distinct co-author countries per article	Higher libdem → higher international collaboration breadth	$\beta > 0$	—	Simple
15	`share_libdem_coauth` Share with $\geq$ 1 co-author from liberal democracy	Higher libdem → higher share co-authored with liberal-democratic partner	$\beta > 0$	—	Medium
16	`share_domestic_only` Share with exclusively domestic authorship	Higher libdem → lower share of domestically-only authored articles	$\beta < 0$	—	Simple
17	`mean_n_authors` Mean author count per article	Higher libdem → higher mean team size	$\beta > 0$	—	Simple

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T	Outcome variable	H1 (condensed)	Sign on $\hat{\beta}$	Extra controls / notes	Type
18	`mean_n_institutions` Mean distinct author institutions per article	Higher libdem → higher institutional diversity per article	$\beta > 0$	`mean_n_coauthor_cntry` (separate mechanism)	Medium
Sub-family: visibility-suppression $k = 4$, $\alpha^* = 0.0125$					
19	`log_cite_ratio_mean` Mean log normalized citation ratio	Higher libdem → higher normalized citation impact	$\beta > 0$	—	Simple
20	`log1p(cite_ratio)` Article-level; interaction with sensitive flag	Citation penalty for autocracy × sensitive topic (positive interaction on libdem × sensitive)	$\beta_{\text{interaction}} > 0$	Field FE; article-level	Medium
21	`prop_ever_cited` Proportion of articles with ≥ 1 citation	Higher libdem → higher share of articles receiving at least one citation	$\beta > 0$	—	Medium
22	`gini_cites` Gini coefficient of `tot_cites` within country-year	Higher libdem → lower citation concentration (Gini)	$\beta < 0$	`log(n_articles)` (mechanical compression)	Simple
Sub-family: ideological-alignment $k = 3$, $\alpha^* = 0.0167$					
23	`share_apolitical` Share NEUTRAL + NONE in 5-label ideology classifier (LLM)	Higher libdem → higher share of liberal/neutral political-economy framing	$\beta > 0$	Regime ideology heterogeneity test; ∼1M abstracts	Complex

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T	Outcome variable	H1 (condensed)	Sign on $\hat{\beta}$	Extra controls / notes	Type
24	`share_legitimating` Share actively legitimizing own political system (LLM)	Higher libdem → lower share of regime-legitimating scholarship	$\beta < 0$	~1M abstracts, $500 ceiling	Complex
25	`share_anti_liberal` Share with anti-liberal-democracy framing (LLM)	Higher libdem → lower share of anti-liberal framing in SSH abstracts	$\beta < 0$	~1M abstracts, $500 ceiling	Complex

Notes: All teams use TWFE (country + year FE, SE clustered by `iso3`) as primary specification and pooled OLS (year FE only) as secondary. Baseline controls: `log(e_gdppc)`, `log(e_wb_pop)`. RC1 = pre-specified robustness check 1 (sample or operationalization). RC2 = replace `v2x_libdem` with `lied_binary` (dichotomous). LLM teams use Claude Haiku API. "Simple" = no API; "Medium" = derived variable; "Complex" = LLM classification.

6 Project Disposition and Current Stage

6.1 Phase 0 — Data preparation (Complete)

- Raw WOS SSH records downloaded and parsed: 3,189,557 rows
- De-duplicated to 2,709,224 unique articles (`wos_id`)
- V-DEM political data merged at `iso3` × `year` level
- Country coverage: 99.98% match rate
- Controls added: log GDP per capita, log population
- Output: `data/agent_corpus.rds` (locked; not tracked in Git)

6.2 Phase 1A — Designer sessions (Complete)

- 25 Designer agent sessions run (Claude Sonnet)
- Each session: read brief + data description, develop RQ within domain
- Output: `rq.md` + `analysis_plan.md` per team
- Uniqueness check enforced: no two teams share outcome + unit + strategy

6.3 Gate B — PI review (Complete, 6 May 2026)

- All 25 teams reviewed by PI
- Gate criteria: clear H1, specified controls, ≥ 2 robustness checks, uniqueness verified, self-censorship framing coherent
- Revisions: Teams 15–25 revised post-review; Team 21 fully redesigned (original interaction design rejected; replaced with `prop_ever_cited`)
- All 25 teams approved

6.4 Pre-registration (Complete, 6 May 2026)

- Script: `scripts/preregister.ps1`
- Creates timestamped `preregistration.md` = RQ + analysis plan
- Git commit: `960525f` ("Pre-registration: 25 teams")
- Repository: GitHub (remote `origin/main`)
- All 25 hypotheses and analysis plans locked

6.5 Phase 1B — Analyst sessions (*Pending*)

- 25 Analyst agent sessions to be run
- Each session: read `preregistration.md`, execute analysis in R, write `primary_results.json` and produce figures
- Simple teams (no API): estimated ~2h per session
- Complex teams (LLM API): estimated ~4–8h per session, $2–500 API budget per team
- Planned order: Simple and Medium teams first (01–04, 07–09, 11, 13–17, 19, 22); Complex teams in parallel (05, 06, 10, 12, 23–25)

6.6 Gate D — PI review of results (*Pending*)

- Review all 25 figures and `primary_results.json` files
- Criteria: analysis executed as pre-registered, results interpretable
- Option to redirect teams with implementation errors only (no outcome-driven changes)

6.7 Phase 2 — Writer sessions (*Pending*)

- 25 Writer agent sessions (8-section standardized report template)
- Bonferroni adjustment computed first: `scripts/bonferroni_adjust.R`

6.8 Gate G — PI review of reports (*Pending*)

6.9 Phase 3 — Reviewer sessions (*Pending*)

- Peer review of each report by a separate Reviewer agent instance

6.10 Phase 4 — Synthesis (*Pending*)

- Synthesizer agent reads all 25 reports + peer reviews
- Output: `synthesis/theory_evaluation.md`
- One verdict per sub-family: Supports / Partially supports / Mixed / Does not support

7 Results

[Section to be written after Analyst sessions complete.]

7.1 Main findings by sub-family

[TODO: Table: Team | Direction | Coefficient | SE | p_raw | p_adj | Sig?]

7.2 Consistency across specifications

[TODO: TWFE vs. pooled OLS comparison.]

7.3 Robustness

[TODO: RC1 and RC2 results summary.]

8 Discussion

[To be written after synthesis.]

Key points (skeleton).

- Which sub-families show most consistent evidence?
- Heterogeneity: where is self-censorship concentrated?
- Limitations: observational, cannot rule out selection, corpus skewed to WOS-indexed output
- Implications for academic freedom research

9 Conclusion

[To be written after analysis.]

A Team roster and file map

Table 3: Team file structure (one row per team)

Team	Sub-family	`rq.md`	`analysis_plan.md`	`preregistration.md`
01–07	topic-avoidance	✓	✓	✓
08–13	framing-neutrality	✓	✓	✓
14–18	collaboration-constraint	✓	✓	✓
19–22	visibility-suppression	✓	✓	✓
23–25	ideological-alignment	✓	✓	✓

B LLM classification prompts

[Full prompts for Teams 05, 10, 12, 23, 24, 25 — to be extracted from analysis plans.]

C Variable codebook

[Key variables from `agent_corpus.rds`. Cross-reference `data/vdem_codebook.md`.]